A whole-cell platform for discovering synthetic cell adhesion molecules in bacteria

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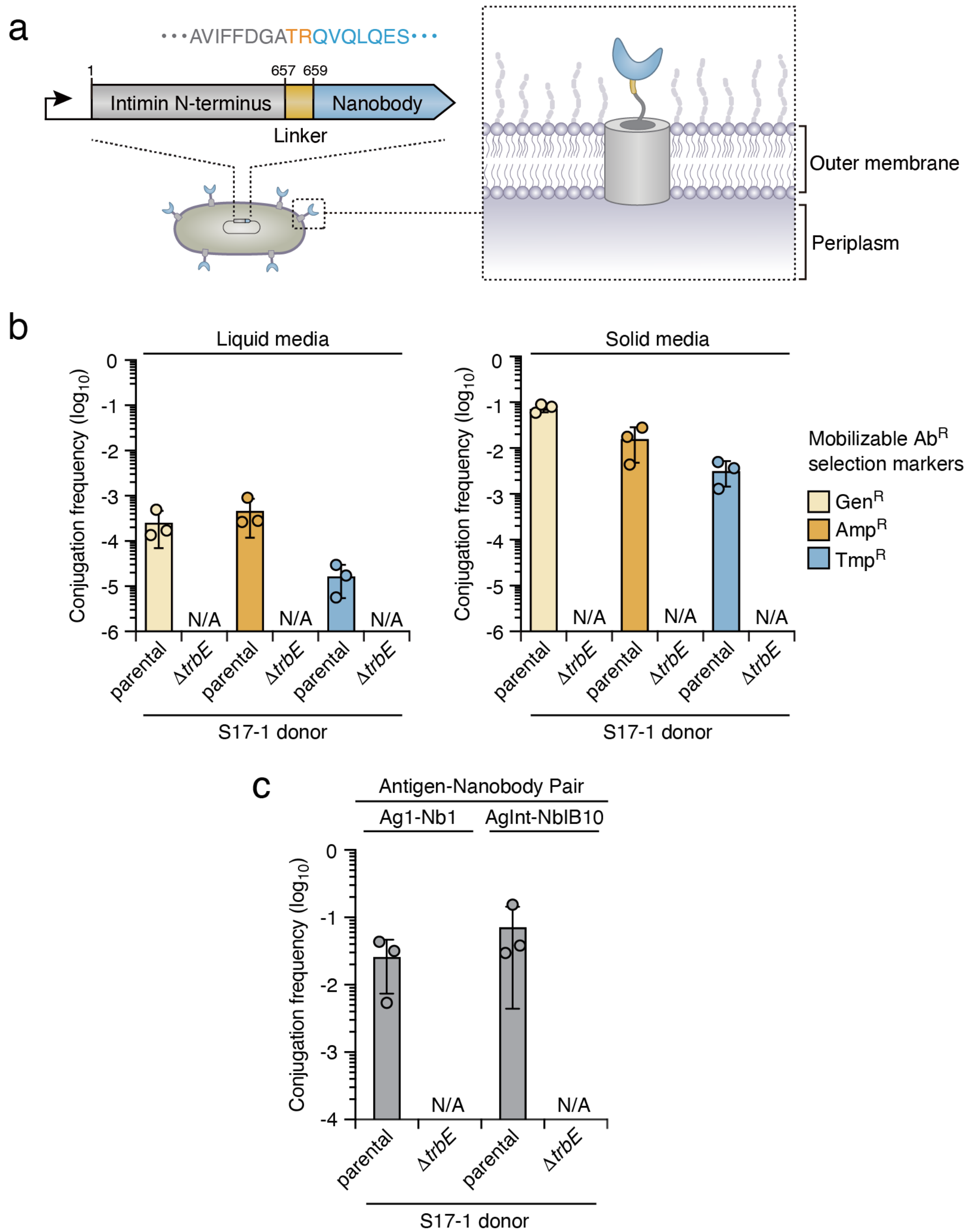
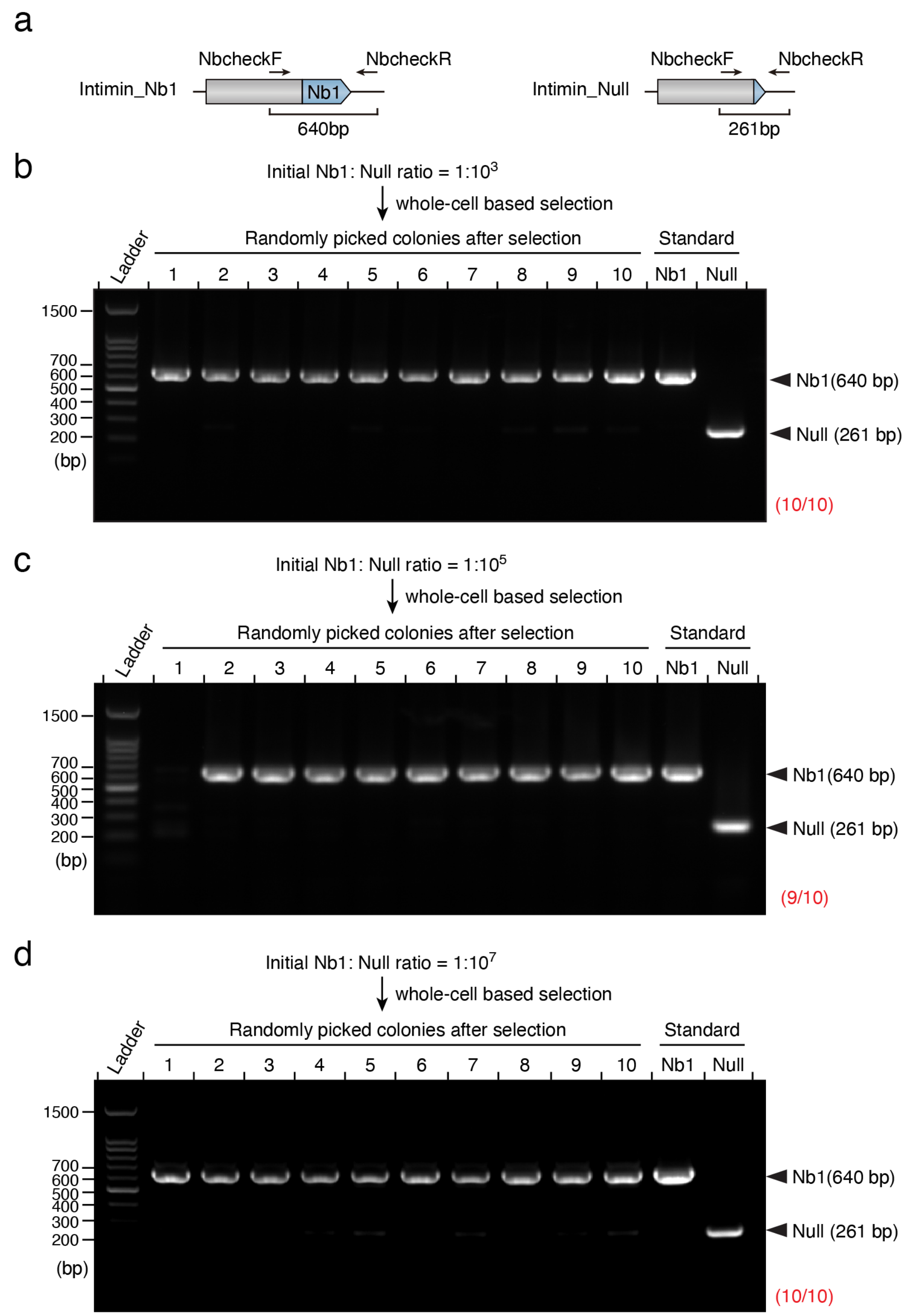
Supplementary Information

The supplementary information consists of 12 Supplementary Figures and 2 Supplementary Tables.

Supplementary Figures



Supplementary Fig. 1: Dependency of selection marker gene transfer on a functional T4SS.
(a) Schematic of the intimin-based display system. The construct contains the N terminus of the autotransporter intimin from enterohemorrhagic *E. coli* (EHEC O157:H7), which serves as an anchor on the outer membrane of the cell. The C terminus of this truncated intimin domain is fused with the nanobody gene, with a linker sequence positioned between them. When expressed in *E. coli*, the intimin N terminus reaches the outer membrane, facilitating presentation of the nanobody in the extracellular space. **(b)** The conjugation frequency of selection markers from the WT and Δ*trbE* (dysfunctional T4SS) donors to recipients. Conjugation assays were performed under both liquid and solid growth conditions, utilizing pGenR, pCarbR, and pTmpR as selection markers. **(c)** The impact of a deficient T4SS on the donor (*E. coli* S17-1) to convey the selection marker (pGenR) to the recipient (*E. coli* MG1655). Two binding pairs were employed: Ag1 and AgInt were displayed by the donors, and their corresponding nanobodies (Nb1 and NbIB10) were displayed by the recipients. Data are presented as means ± SD. n = 3 biological replicates with three technical replicates each. Conjugation frequency lower than the detection limit is designated as N/A. Source data are provided as a Source Data file.



Supplementary Fig. 2: Diagnostic PCR verification of the efficacy of the selection strategy in enriching for low-abundance nanobodies.
(a) The display system employed in this study comprises the intimin N-terminus (grey) fused with the adhesin region (yellow) at the C-terminus. Arrows indicate the annealing primers utilized for diagnostic PCR. The expected length of the PCR products for the cells displaying Nb1 (640 bp) and null (261 bp) are underscored. **(b)** PCR analysis of the transconjugants after selection. Two recipient strains expressing Nb1 and non-matched null control, respectively, were mixed at an initial ratio of $1{:}10^3$. The primers illustrated in **(a)** were used for diagnostic PCR. The proportions of colonies expressing Nb1 are denoted at the bottom-right (red). **(c)** Randomly selected colonies after two rounds of bio-panning were subjected to diagnostic PCR. The recipient cells displaying Nb1 were diluted with cells expressing a non-matched null control at an initial ratio of $1{:}10^5$. **(d)** The recipient population, initially at a ratio of Nb1:Null = $1{:}10^7$, after three rounds of selection was verified through diagnostic PCR.

a

Top 10 Nb^{traN} candidates

```
    1                                CDR1                CDR2          60
1.  QVQLQESGGGLVQAGGSLRLSCAASGTISDAGYMGWYRQAPGKERELVASISAGGITYYADSVKGR
2.  QVQLQESGGGLVQAGGSLRLSCAASGTISLDYGMGWYRQAPGKEREFVATISYGANTNYADSVKGR
3.  QVQLQESGGGLVQAGGSLRLSCAASGTIFDGVDMGWYRQAPGKEREFVAGITAGSSTYYADSVKGR
4.  QVQLQESGGGLVQAGGSLRLSCAASGTISVGTYMGWYRQAPGKEREFVAGISYGSITNYADSVKGR
5.  QVQLQESGGGLVQAGGSLRLSCAASGTIFTTVVMGWYRQAPGKEREFVAGIDWGTNTNYADSVKGR
6.  QVQLQESGGGLVQAGGSLRLSCAASGNIFYHGVMGWYRQAPGKEREFVAAITRGTNTNYADSVKGR
7.  QVQLQESGGGLVQAGGSLRLSCAASGTIFFYGRMGWYRQAPGKEREFVATIGYGASTYYADSVKGR
8.  QVQLQESGGGLVQAGGSLRLSCAASGNIFWVVTMGWYRQAPGKEREFVAGIAQGGNTYYADSVKGR
9.  QVQLQESGGGLVQAGGSLRLSCAASGNIFAGDGMGWYRQAPGKEREFVASINIGASTYYADSVKGR
10. QVQLQESGGGLVQAGGSLRLSCAASGTISYVYDMGWYRQAPGKERELVASIGRGTTTYYADSVKGR

   61                                        CDR3
1.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAVGPDGTDSHGYWGQGTQVTVSS*
2.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAAYQPHTGSLLYWGQGTQVTVSS*
3.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAAYPLDEIGSHDPHSYWGQGTQVTVSS*
4.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAARRYRSDPSYQYHVYWGQGTQVTVSS*
5.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAVTGSVYVDHRYWGQGTQVTVSS*
6.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAVRKIWDRYYNYWGQGTQVTVSS*
7.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAARCFLRFVFSYWGQGTQVTVSS*
8.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAVNGWGYSFIVWDSLYYWGQGTQVTVSS*
9.  DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAAYWDYAYYTDKARNRYYTYWGQGTQVTVSS*
10. DSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAVEQLDWDGSSPYPNTILIYWGQGTQVTVSS*
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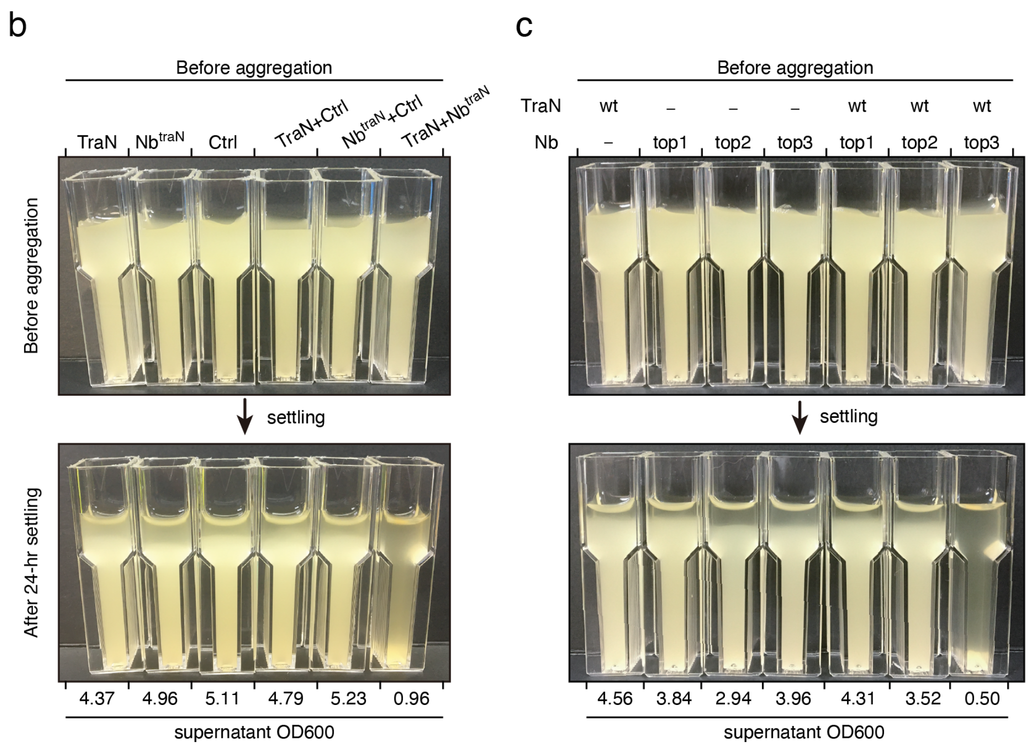
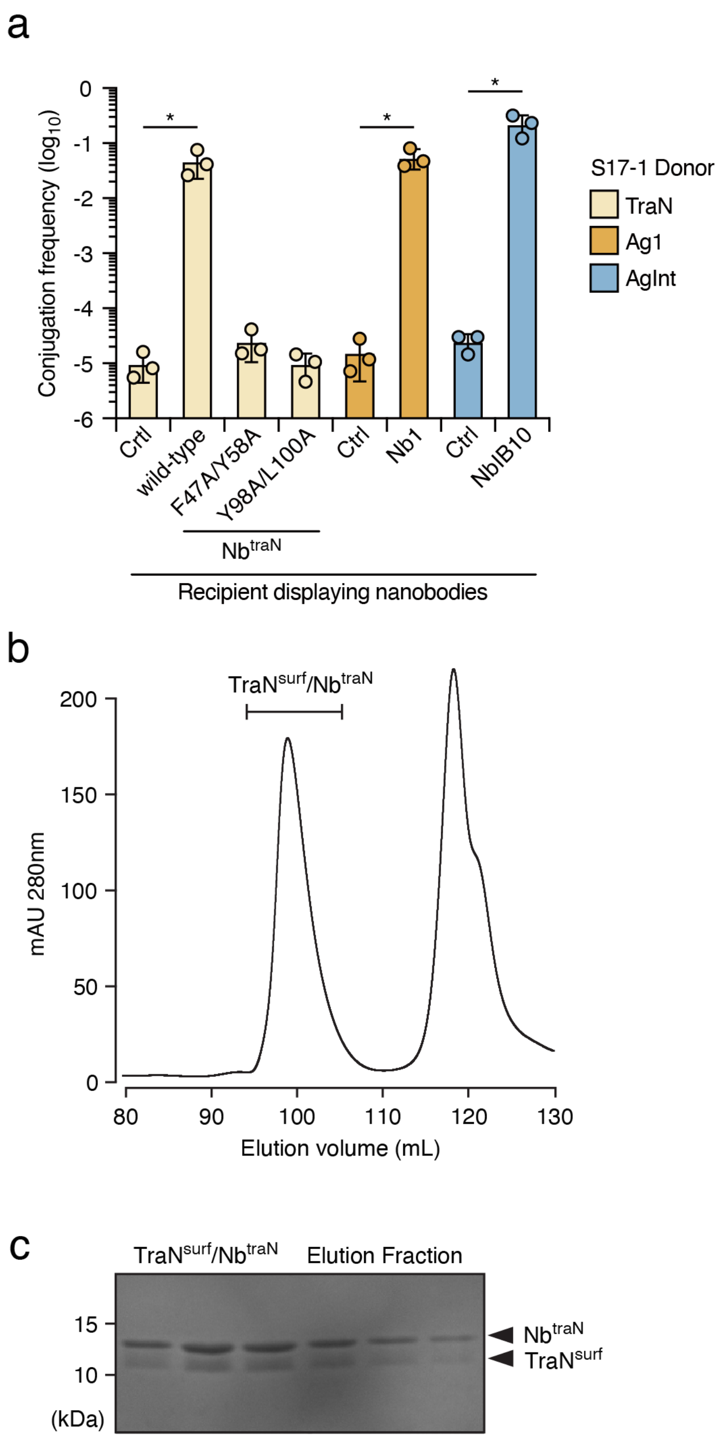
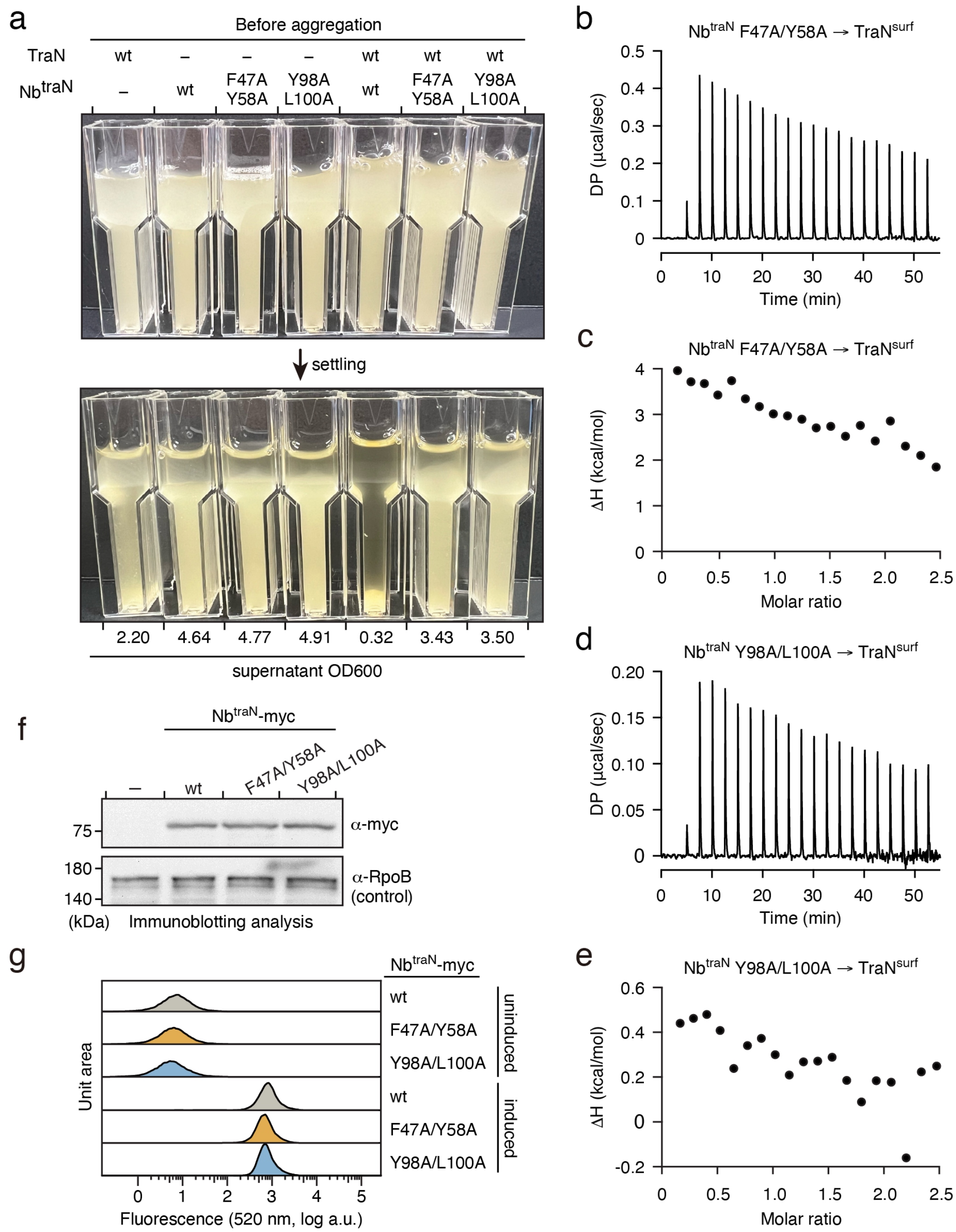


Supplementary Fig. 3: TraN-targeted nanobodies identified from the screening.
(a) Amino acid sequences of Nb^{traN} candidates. The top 10 nanobody sequences that were most enriched after selection are listed. Each sequence shows the CDR1, CDR2, and CDR3 in green, blue, and orange, respectively. **(b)** Macroscopic aggregation analysis to assess cell-cell adhesion of the indicated cultures of *E. coli* strains either individually or in combination. *E. coli* displaying TraN, Nb^{traN}, or the non-matched control (Ctrl) were tested. The OD_{600} of the culture supernatants was measured after the settling period. **(c)** Macroscopic aggregation analysis to assess cell-cell adhesion between *E. coli* strains displaying TraN and the top three Nb^{traN} candidates shown in **(a)**. The strains were incubated either individually or in combination. The OD_{600} of the culture supernatant is shown below the image.



Supplementary Fig. 4: Nb^{traN} binds to TraN.
(a) Conjugation frequency of pGenR under the liquid growth condition. Recipient cells expressing indicated nanobodies or control (Ctrl, without nanobody displayed) were co-cultured with donor cells expressing TraN, Ag1, or AgInt. Data are presented as means ± SD. n = 3 biological replicates with three technical replicates each, ± SD. Asterisks indicate statistically significant differences between the indicated mean values ($p<0.05$, two-tailed t-test). **(b)** Size-exclusion chromatography profile of the Nb^{TraN}/TraN^{surf} complex using a HiLoad 16/600 Superdex 200pg column. **(c)** The Coomassie-stained SDS-PAGE protein gel shows the elution fractions of Nb^{traN} in complex with TraN^{surf}. Migrations of a size standard, in kDa, are indicated (left). Source data are provided as a Source Data file.



Supplementary Fig. 5: Mutations in the CDR2 or CDR3 of Nb^{traN} affect its interaction with TraN.

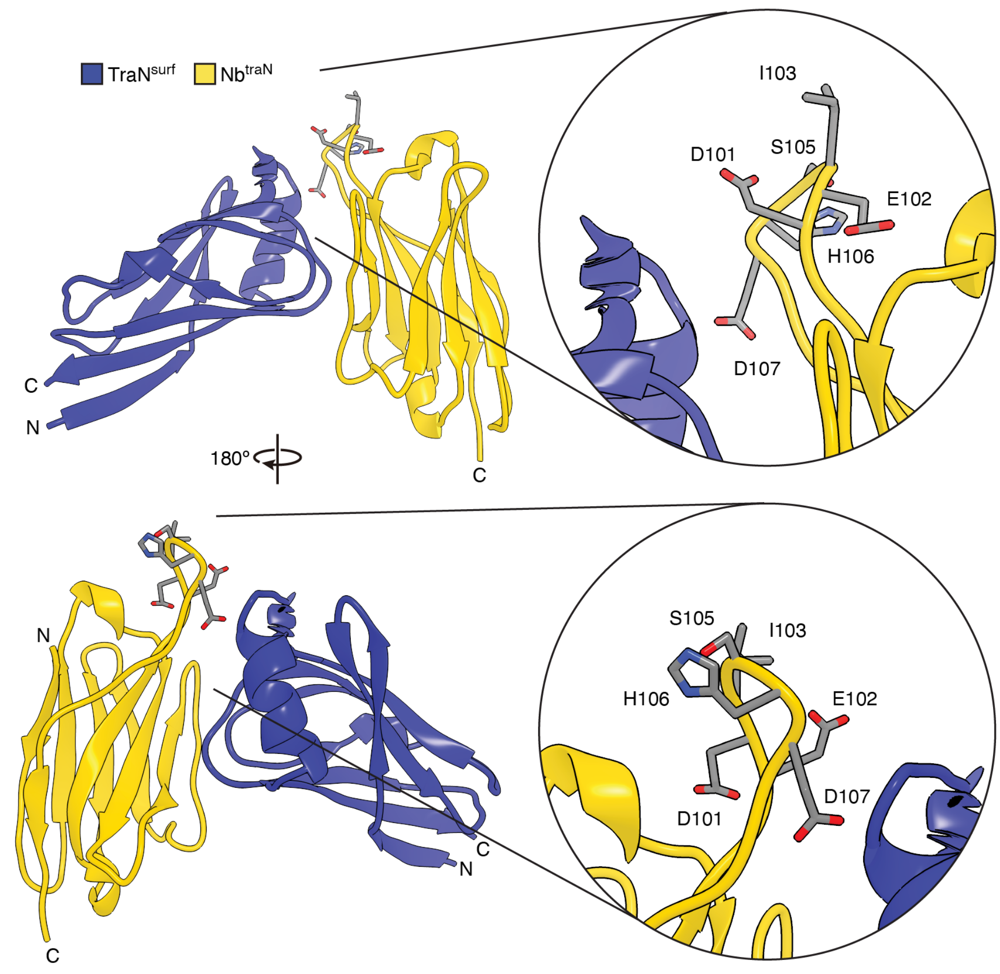
(a) Comparison of the adhesive capability of the wild-type (wt) Nb^{traN} and its variants (CDR2: F47A/Y58A; CDR3: Y98A/L101A). For the macroscopic aggregation analysis, single cultures and mixtures were used. After settling, the OD_{600} of the culture supernatants was measured and values are shown below the image. (**b, c**) The ITC binding profiles of Nb^{traN} F47A/Y58A with TraN^{surf}. **(d, e)** The ITC binding profiles of Nb^{traN} Y98A/L101A with TraN^{surf}. **(f)** Immunoblotting analysis of *E. coli* expressing the myc-tagged intimin-Nb^{traN} variants (wild-type, F47A/Y58A, and Y98A/L100A) after a 16-hour incubation. *E. coli* without expressing nanobody was used as a control (-). The loading control is *E. coli* RNA Polymerase β protein (RpoB). Migrations of a size standard, in kDa, are indicated (left). **(g)** Immunofluorescence and flow cytometry analysis of *E. coli* expressing myc-tagged Nb^{traN} or its variants (F47A/Y58A and Y98A/L100A) on the cell surface. Cells were treated with anti-myc antibodies followed by Alexa fluor 488-conjugated secondary antibody. The X-axis represents the fluorescence intensity at 525/40 nm and the Y-axis indicates the cell counts normalized by unit area. Source data are provided as a Source Data file.

a

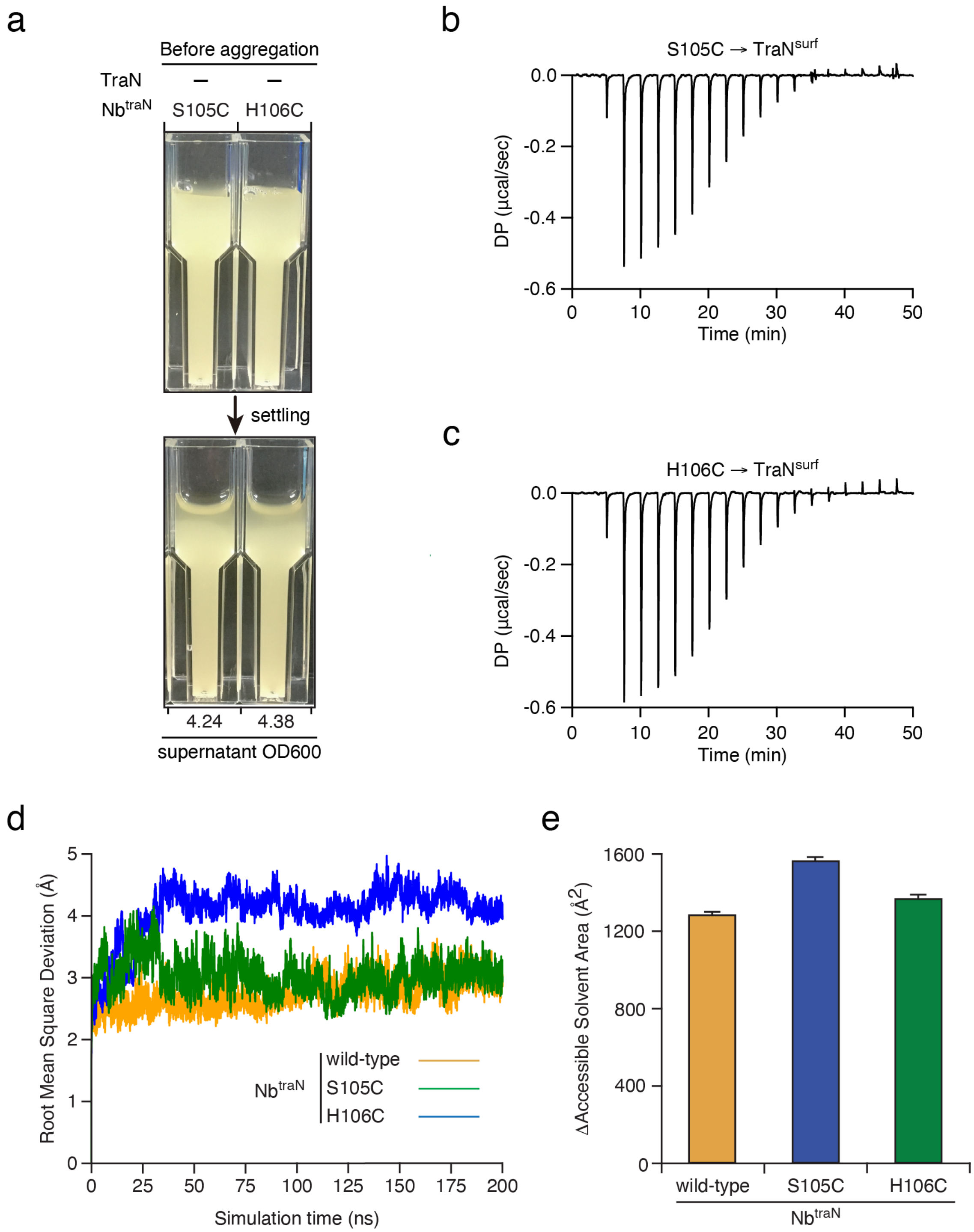
		aliphatic					aromatic			polar						positive			negative	
CDR3 amino acid position	Mutation:	G	P	A	I	L	V	F	W	Y	M	C	S	T	N	Q	H	K	R	D	E
A	97	-2.00	-9.25	0.00	-5.59	-3.31	-5.31	-3.89	-6.00	-4.90	-3.87	-1.86	-3.44	-5.67	-9.18	-8.16	-8.11	-8.33		-9.66	
Y	98	-8.63	-8.10	-8.96		-9.15	-9.16	-1.75	-8.21	0.00		-7.02	-6.91	-9.21	-7.64		-3.51		-8.49	-7.90	-9.07
P	99	-7.08	0.00	-4.73	-3.72	-5.08	-3.01	-4.45	-2.84	-4.48	-5.04	-3.94	-6.72	-5.85		-5.67	-5.13		-9.16	-3.89	-4.10
L	100	-6.45	-8.37	-5.83	-0.50	0.00	-3.36	-0.39	0.82	-0.54	0.48	-3.45	-6.18	-4.74	-5.86	-3.56	-5.90	-5.75	-4.75	-7.39	-7.48
D	101	-1.13	-1.28	-0.21	-0.22	-0.38	-0.24	-2.09	-6.37	-0.26	-0.55	0.94	-0.49	-0.44	0.21	-0.52	-0.25	0.51	-1.41	0.00	0.06
E	102	1.27	0.93	0.33	0.15	0.45	0.67	-0.04	-4.23	0.20	0.86	2.16	0.57	1.03	1.30	0.23	0.54	0.10	-0.58	0.56	0.00
I	103	0.34	0.88	0.01	0.00	-0.07	0.06	-1.21	-1.51	-0.82	-0.06	2.27	0.15	0.27	-0.53	0.40	0.01	0.07	-1.29	-0.03	0.17
G	104	0.00	0.52	0.01	0.08	0.73	0.39	-0.44	-4.42	0.36	-0.37	3.15	0.13	0.24	0.40	0.16	0.14	-0.37	-0.80	0.57	0.85
S	105	1.15	-0.34	-0.14	-0.72	-0.98	0.19	-7.15	-8.06	-0.69	-0.99	3.20	0.00	0.46	0.04	-0.06	-0.43	-0.49	-1.85	0.82	0.54
H	106	0.81	-0.92	0.80	0.67	0.24	1.19	-0.41	0.17	0.57	0.48	4.13	0.83	0.79	0.65	0.67	0.00	0.66	-0.20	1.14	1.46
D	107	-4.71	-4.35	-5.66			-7.05			-5.27		1.27	-5.71	-6.60	-4.48	-8.09	-7.86	-7.69		0.00	-3.86
P	108		0.00	-5.12	-4.07	-6.82	-3.66	-6.73	-8.09	-6.59	-8.06	-0.69	-6.89	-6.18		-5.78	-1.72			-7.02	-7.73
H	109	-5.04	-7.05	-2.81	-0.08	-2.63	-0.37	-3.21	-4.60	-3.45	-0.83	-0.49	-2.57	-1.28	-2.91	0.16	0.00	-0.40	-0.16	-1.92	-0.69
S	110	-1.11	0.33	1.69	-5.07	-8.36	-0.75		-6.88	-8.52	-6.61	-1.38	0.00	0.54	-6.76	-4.17	-8.09	-7.17	-8.69	0.83	2.01

Conjugative score: +5; 0 (Nb^{traN}); -10; N/A

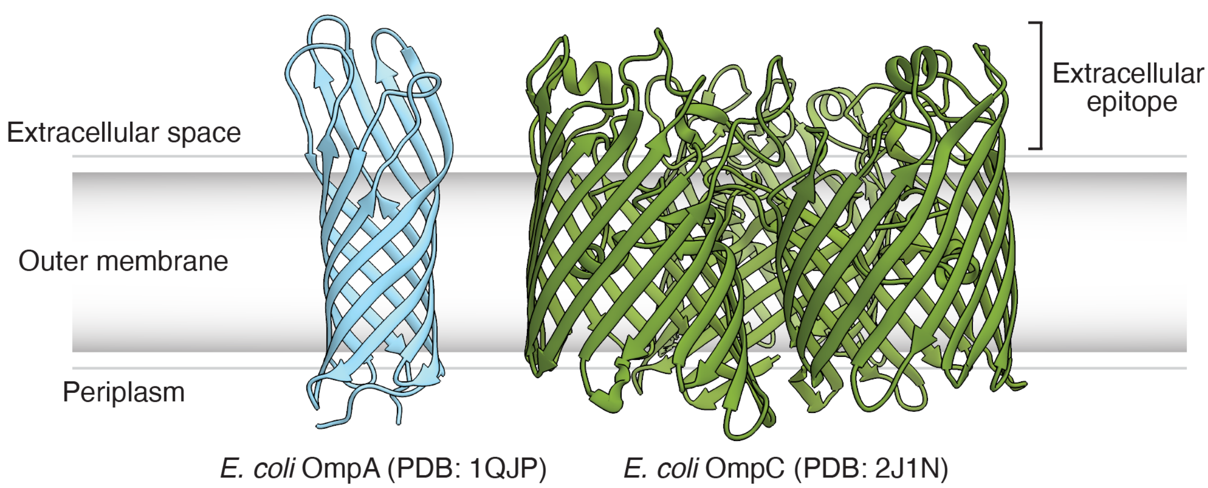
b



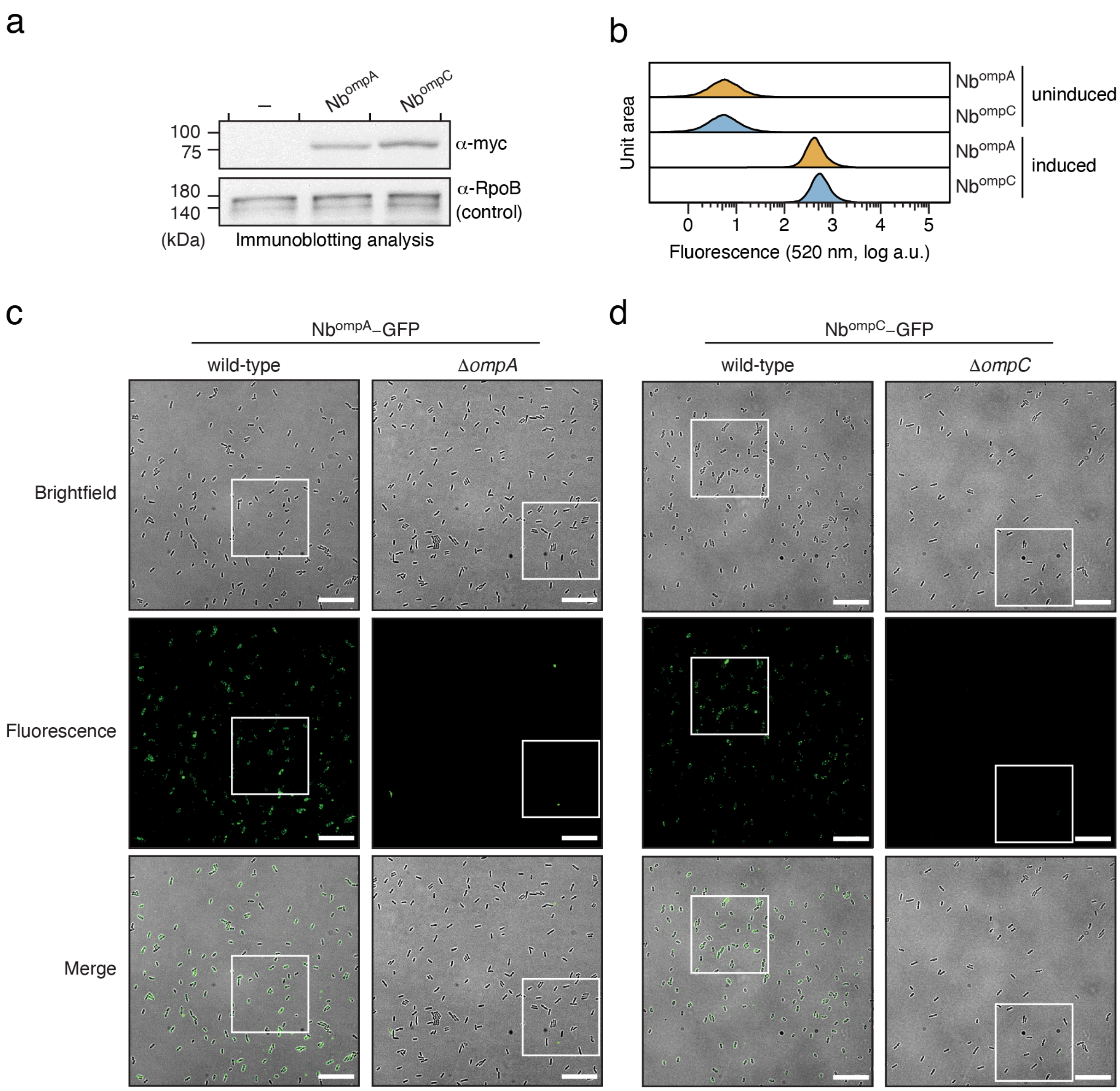
Supplementary Fig. 6: Deep mutational engineering of CDR3 in Nb^{traN}.
(a) The values of the conjugative scores of single amino acid substitution within CDR3 of Nb^{traN}. Related to **Fig. 4**. (**b)** Crystal structure of Nb^{traN} (yellow) bound to the surface-exposed domain of TraN ($TraN^{surf}$, blue) (PDB: 8X7N). A second view (rotated 180° around the vertical axis) is shown below. The close-up views on the right side indicate the locations of the cysteine variants (residues 101 to 107) on Nb^{traN}. Source data are provided as a Source Data file.



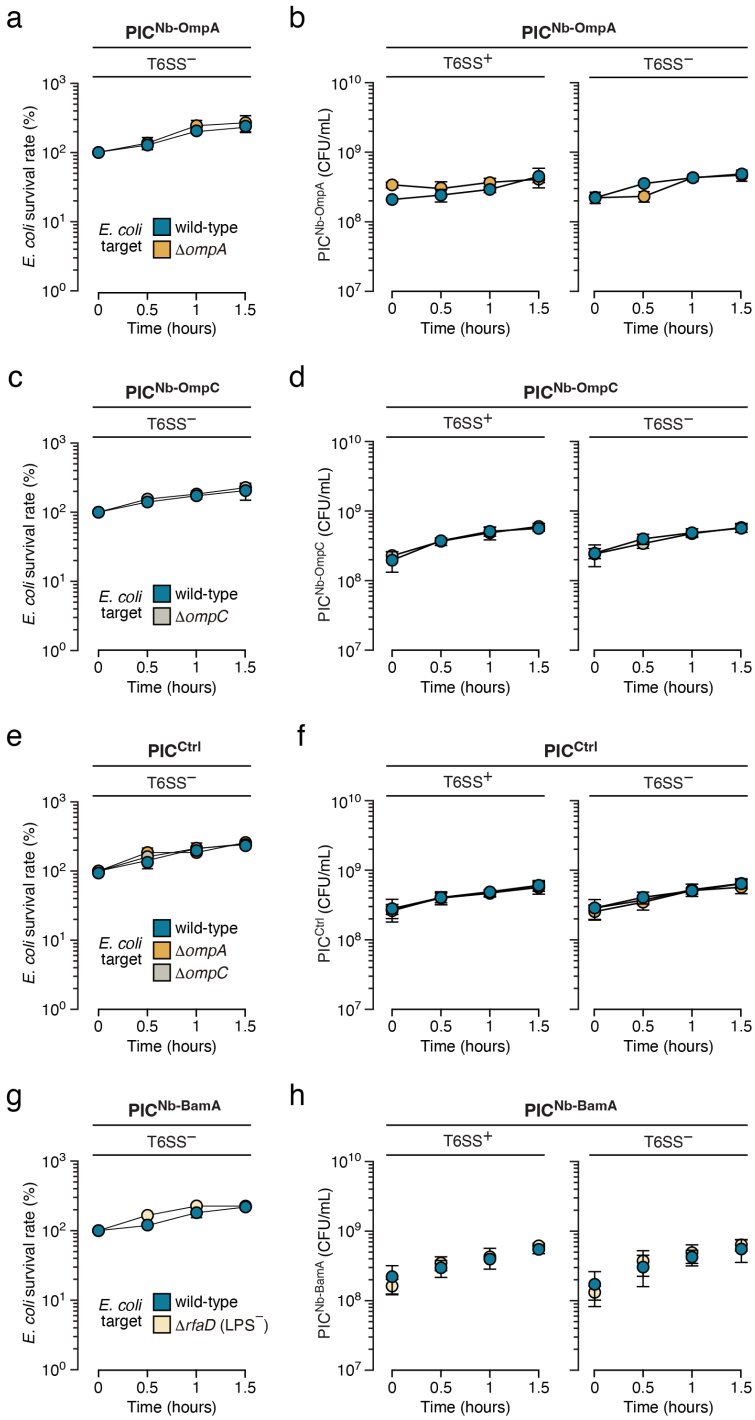
Supplementary Fig. 7: Residue substitutions of CDR3 in Nb^{traN} improve its binding to TraN.
(a) Control cultures of *E. coli* strains expressing the Nb^{traN} variants that were used in the macroscopic aggregation assay in **Fig. 4**. **(b, c)** The ITC binding profiles of Nb^{traN} S105C (**b**) and H106C **(c)** with TraN^{surf}. Related to **Fig. 4**. **(d)** RMSD plots of 200 ns molecular dynamic simulations of Nb^{traN} variants. **(e)** Changes in the accessible solvent area (ΔASA) between the complexes (TraN with Nb^{traN} variants: wild-type, S105C, or H106C) and their constituent proteins were simulated independently. Data are presented as means with 95% CI. Source data are provided as a Source Data file.



Supplementary Fig. 8: Structures of *E. coli* OmpA and OmpC.
Crystal structures of the OmpA membrane domain (PBD: 1QJP)[1] and OmpC (PDB: 2J1N)[2] from *E. coli*. OmpA (blue) is presented as a monomeric structure, featuring an eight-stranded β-barrel spanning the outer membrane, coupled with flexible loops at its external end. Homotrimeric OmpC (green) is composed of three beta-barrels. The monomeric form of OmpC exhibits a 16-stranded β-barrel porin fold, with irregular extracellular loops.

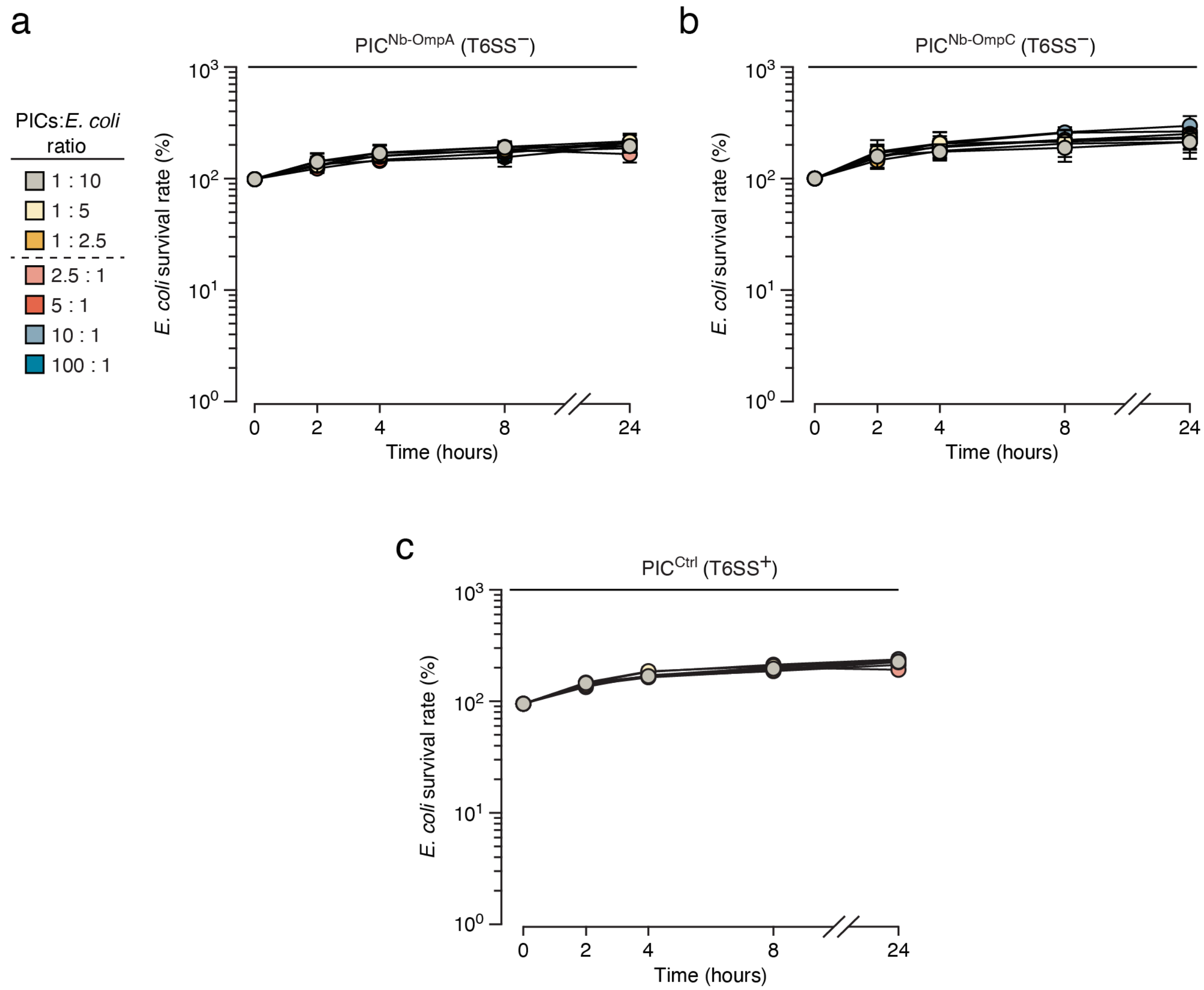


Supplementary Fig. 9: Nb^{ompA}-GFP and Nb^{ompC}-GFP bind to their target antigens.
(a) Immunoblotting analysis of *E. coli* expressing Nb^{ompA} or Nb^{ompA} carrying a C-terminal myc tag after a 16-hour incubation. *E. coli* not expressing nanobody was used as a control (-). The loading control is *E. coli* RNA Polymerase β protein (RpoB). Migrations of a size standard, in kDa, are indicted (left). **(b)** Cell surface accessibility assays of *E. coli* expressing myc-tagged intimin-nanobody fusion proteins. Cells were treated with anti-myc antibodies followed by Alexa fluor 488-conjugated secondary antibody. The X-axis represents the fluorescence intensity at 525/40 nm and the Y-axis indicates the cell counts normalized by unit area. **(c)** The phase-contrast (top), green fluorescence (middle), and merged (bottom) images of *E. coli* (wild-type and Δ*ompA* strain) following 1-hour incubation with Nb^{ompA}-GFP. The white borders demarcate the cropped images displayed in **Fig. 5e**. **(d)** Fluorescence micrographs of *E. coli* (wild-type and Δ*ompC* strain) cells treated with Nb^{ompC}-GFP for 1 hour. Phase-contrast (top), green fluorescence (middle), and merged (bottom) images are presented. Scale bar = 5 µm. Zoomed-in images of the areas within white borders are featured in **Fig. 5f**. Source data are provided as a Source Data file.

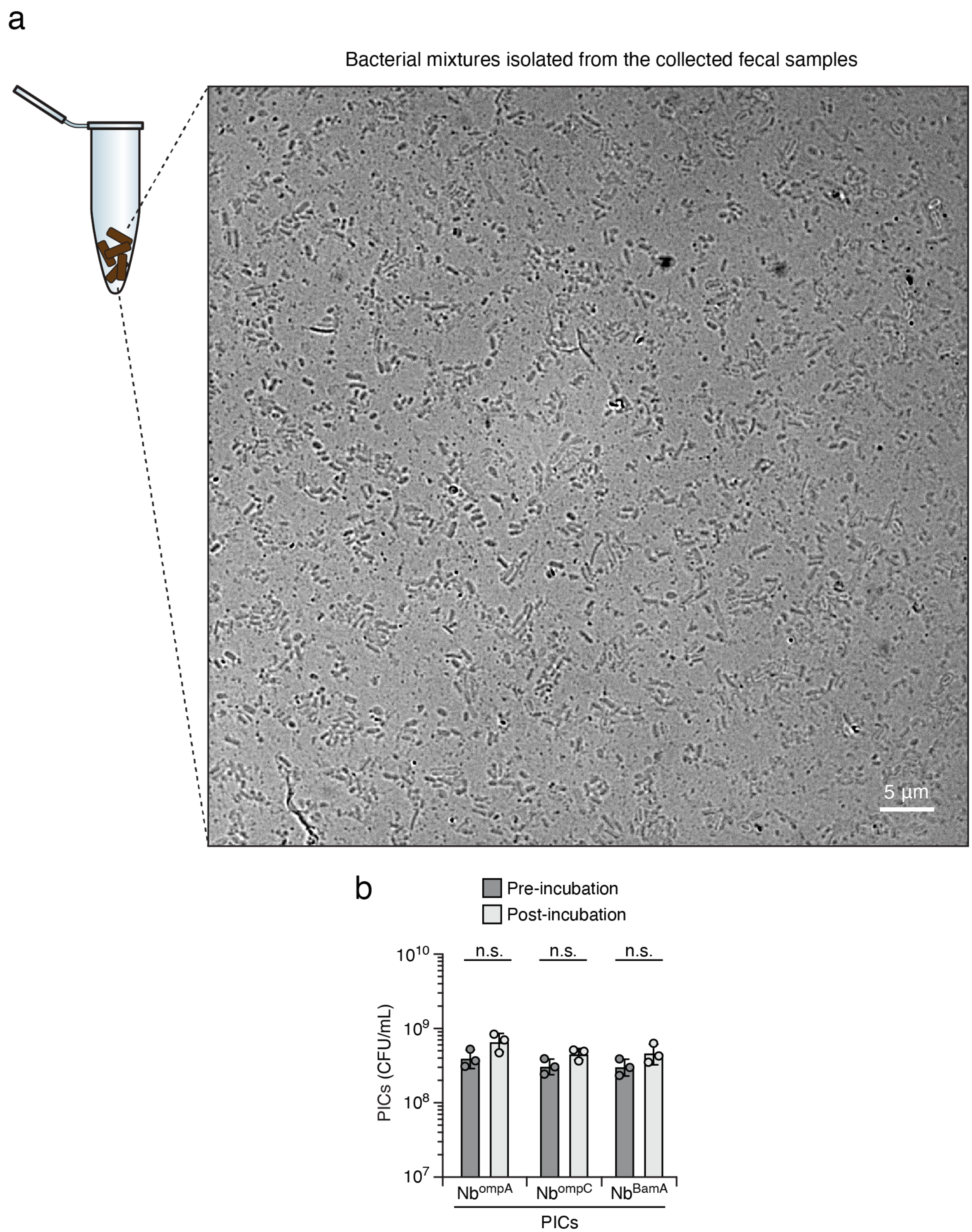


Supplementary Fig. 10: Specific killing mediated by PICs requires a functional T6SS, and survival of PICs upon incubation with target *E. coli*.

(a, c, e, g) Survival rate of *E. coli* wild-type, Δ*ompA*, Δ*ompC*, and Δ*rfaD* strains upon liquid co-culture with T6SS-deficient PICs expressing $\mathrm{Nb}^{\mathrm{ompA}}$ **(a)**, $\mathrm{Nb}^{\mathrm{ompC}}$ **(c)**, non-matched control **(e)**, or $\mathrm{Nb}^{\mathrm{BamA}}$ **(g)**. **(b, d, f, h)** The survival (CFUs/mL) of PICs expressing $\mathrm{Nb}^{\mathrm{ompA}}$ **(b)**, $\mathrm{Nb}^{\mathrm{ompC}}$ **(d)**, non-matched control **(f)**, or $\mathrm{Nb}^{\mathrm{BamA}}$ **(h)** upon liquid co-culture with *E. coli* strains (wild-type, Δ*ompA*, Δ*ompC*, and Δ*rfaD* cells). PICs with a functional (T6SS+) or dysfunctional T6SS (T6SS−, Δ*icmF*) were utilized. Data are presented as means ± SD. n = 3 biological replicates with three technical replicates each. Asterisks indicate statistically significant differences between the indicated mean values ($p<0.05$, two-tailed t-test). Source data are provided as a Source Data file.



Supplementary Fig. 11: PICs lacking a functional T6SS do not influence the survival rate of *E. coli*.
(a, b, c) Survival rate of wild-type *E. coli* during 24-hour incubation with PICs expressing Nb^{ompA} **(a),** Nb^{ompC} **(b),** or Null **(c)**. PICs with a dysfunctional T6SS (T6SS−, Δ*icmF*) were used in **(a, b)**. PICs with an active T6SS (T6SS+) were used in **(c)**. Different starting ratios of PICs to *E. coli* were tested, including 1:10, 1:5, 1:2.5, 2.5:1, 5:1, 10:1, and 100:1. Data are presented as means ± SD. n = 3 biological replicates with three technical replicates each. Asterisks indicate statistically significant differences between the indicated mean values ($p<0.05$, two-tailed t-test). Source data are provided as a Source Data file.



Supplementary Fig. 12: Survival of PICs remains unaffected before and after incubation in a bacterial mixture derived from mice fecal samples.
(a) Phase-contrast micrograph of a bacterial population isolated from mouse fecal samples. Scale bar = 5 µm. (**b**) Survival (CFUs/mL) of PICs before and after incubation with mouse fecal microbiota. Related to **Fig. 6j**. Data are presented as means ± SD. n = 3 biological replicates with three technical replicates each. Asterisks indicate statistically significant differences between the indicated mean values ($p<0.05$, two-tailed t-test). Source data are provided as a Source Data file.

Supplementary Table 1. X-ray data collection and refinement statistics. Relevant to Method and Fig. 3.

Crystals	$TraN^{surf}/Nb^{traN}$ complex
Space group	$2_1\ 2_1\ 2_1$
Cell dimensions	
a, b, c (Å)	a= 96.864 b=153.263 c=163.03
α, β, γ (o)	α=β=γ=90
Data collection	
Wavelength (Å)	0.97625
Resolution	19.85 -3.67
Total reflections	230515
Unique reflections	26580
R_{merge} (%)[a]	20.5 (63.9)
CC1/2 (%)[a]	98.8 (95.5)
$I / \sigma I$ [a]	7.3 (2.6)
Completeness (%)	99.8 (98.5)
Redundancy	8.6
Refinement	
R_{work} / R_{free} (%)	25.96/28.54
Average B-factor	84.2
Bond length rmsd (Å)	0.002
Bond angle rmsd (°)	0.478
Ramachandran plot (%)	
Favored region	94.10
Outlier region	0.0
PDB code	8X7N

[a] Statistics for the highest-resolution shell are shown in parentheses.

Supplementary Table 2. Strains, Recombinant DNA, and Oligonucleotides used in this study.

Strain	Source	Identifier
Escherichia coli S17-1 ATCC 47055	American Type Culture Collection	ATCC 47055
Escherichia coli S17-1 ATCC 47055 Δ*trbE*	This paper	N/A
Escherichia coli MG1655 Tn7::Cat	This paper	N/A
Escherichia coli MG1655 Tn7::Gen^R	This paper	N/A
Escherichia coli MG1655 Δ*ompA* Tn7::Cat	This paper	N/A
Escherichia coli MG1655 Δ*ompC* Tn7::Cat	This paper	N/A
Escherichia coli MG1655 Δ*ompW* Tn7::Cat	This paper	N/A
Escherichia coli MG1655 Δ*rfaD*::Cat	Ting et al., 2020[3]	N/A
Enterobacter cloacae ATCC 13047	American Type Culture Collection	ATCC 13047
Enterobacter cloacae ATCC 13047 Δ*icmF*	Whitney et al., 2014[4]	N/A
Salmonella enterica subsp. *enterica* serovar Typhimurium str. LT2	Gift from Dr. Shu-Jung Chang	N/A
Escherichia coli DH5α	YB Biotech	Cat# XR-LYE678-80
Escherichia coli S17-1 λ pir	Lab collection	N/A
Escherichia coli BL21 pRIL (DE3)	Gift from Dr. Hung-Ta Chen	N/A
Yeast-Display Nanobody Library (NbLib)	Kerafast[5]	Cat#EF0014-FP
E. coli-Display Nanobody Library	This paper	N/A

Recombinant DNA	Source
pDSG_Null (pDSG323)	Glass and Riedel-Kruse, 2018[6]
pDSG_Nb1 (pDSG372)	Glass and Riedel-Kruse, 2018[6]
pDSG_NbIB10 (pDSG_Nb-Int)	Ting et al., 2020[3]
pDSG_Nb-BamA	Ting et al., 2020[3]
pDSG_Nb-TraN	This paper

pDSG_Nb-TraN-myc	This paper
pDSG_Nb-TraN_F47A/Y58A	This paper
pDSG_Nb-TraN_Y98A/L100A	This paper
pDSG_Nb-TraN_F47A/Y58A-myc	This paper
pDSG_Nb-TraN_Y98A/L100A-myc	This paper
pDSG_Nb-TraN_S105C	This paper
pDSG_Nb-TraN_S105C-myc	This paper
pDSG_Nb-TraN_H106C	This paper
pDSG_Nb-TraN_H106C-myc	This paper
pDSG_Nb-OmpA	This paper
pDSG_Nb-OmpC	This paper
pDSG_Nb-OmpA-myc	This paper
pDSG_Nb-OmpC-myc	This paper
pDSG_Ag1 (pDSG358)	Glass and Riedel-Kruse, 2018[6]
pDSG_AgInt (Intimin full length)	Ting et al., 2020[3]
pDSG_TraN	This paper
pET-Duet-1-His6_TraN^{surf}	This paper
pET-Duet-1-His6_TraN^{surf} / Nb-TraN	This paper
pET22b_Nb-TraN_His6	This paper
pET22b_Nb-TraN F47A/Y58A_His6	This paper
pET22b_Nb-TraN Y98A/L101A_His6	This paper
pET22b_Nb-TraN S105C_His6	This paper
pET22b_Nb-TraN H106C_His6	This paper
pET22b_Nb-OmpA_His6	This paper
pET22b_Nb-OmpC_His6	This paper
pET11a-GFP-avi	Gift from Jen-Hsuan Wei
pET22b_Nb-OmpA-GFP_His6	This paper
pET22b_Nb-OmpC-GFP_His6	This paper
pBAD18_OmpA	This paper
pBAD18_OmpC	This paper
pRE118_EcoliΔ*ompW*	This paper
pGenR	This paper
pAmpR	Choi et al., 2005
pTmpR	This paper
pKD4	Datsenko and Wanner, 2000[7]
pKD46	Datsenko and Wanner, 2000[7]
pCP20	Cherepanov and Wackernagel, 1995[8]

Primer	**Sequences (5'-3')**	**Identifier**
Library construction		
Yeast_Nb_F	TTATATTTTTGATGGTGCG ACTAGACAGGTGCAGCTGC AGGAAAGCGGC	Mission Biotech

Yeast_Nb_R	CCCTTTTTGCCGGACTGCAGTTATTAGCTGCTCACGGTCACCTGGGTGCC	Mission Biotech
pDSG_Null_backbone_F	CTAATAACTGCAGTCCGGCAAAAAAGGGCAAG	Mission Biotech
pDSG_Null_backbone_R	TGTCTAGTCGCACCATCAAAAAATATAACCGC	Mission Biotech
Bacteria strain construction		
EcoliS17_ΔtrbE_F	CGCTCGACCCCGTTCCGCGAGAACACCAATAGCCAAGGGAAGCAATACCGTGTAGGCTGGAGCTGCTTCG	Mission Biotech
EcoliS17_ΔtrbE_R	CGTTGCGGGCTTCTTCTTGAAGATCAAGCCCTTGATCGTGTCTGCAAAACCATATGAATATCCTCCTTA	Mission Biotech
TrbE_seqF	GTTCCGCGAGAACACCAATA	Mission Biotech
TrbE_seqR	GCGGGCTTCTTCTTGAAGAT	Mission Biotech
MG1655_ΔompW_F1	CATGAATTCCCGGGAGAGCTCCATGATGCCAATCTGTAC	Mission Biotech
MG1655_ΔompW_R1	ATTAAAAACGCTTTTTCATATCCGCTCC	Mission Biotech
MG1655_ΔompW_F2	TATGAAAAAGCGTTTTTAATTCCGCACAAAAAC	Mission Biotech
MG1655_ΔompW_R2	CGACGGATCCCAAGCTTCTTCTAGAAAAGTCGAGCATTATGAGATTG	Mission Biotech
MG1655_ΔompW_seq_F2	GTCCTGTACCAACATCACCG	Mission Biotech
MG1655_ΔompW_seq_R2	AGGTCATCGAAAGTACCGAG	Mission Biotech
MG1655_ΔompA_F	GAGGCTTGTCTGAAGCGGT	Mission Biotech
MG1655_ΔompA_R	GCGGAACACCAGCATACGC	Mission Biotech
MG1655_ΔompA_seq_F	CAATGCTTTCAGTCGTGAC	Mission Biotech
MG1655_ΔompA_seq_R	AACCATTCGGGCAAGACG	Mission Biotech
MG1655_ΔompC_F	GTTGCGAGTGAAGGTTTTGTTTTGAC	Mission Biotech
MG1655_ΔompC_R	GCGTCCAGTTGGGTCTGAATC	Mission Biotech
MG1655_ΔompC_seq_F	GAGGCATCCGGTTGAAATAG	Mission Biotech
MG1655_ΔompC_seq_R	GCGATAAACTTTGCGAGTCG	Mission Biotech

Plasmid construction		
pBAD18_MG1655_ompA_F	TGGGCTAGCGAATTCGAGCTCATGAAAAAGACAGCTATCGCG	Mission Biotech
pBAD18_MG1655_ompA_R	TGCATGCCTGCAGGTCGACTCTAGATTAAGCCTGCGGCTGAGTTAC	Mission Biotech
pBAD18_MG1655_ompC_F	TGGGCTAGCGAATTCGAGCTCATGAAAGTTAAAGTACTGTCC	Mission Biotech
pBAD18_MG1655_ompC_R	TGCATGCCTGCAGGTCGACTCTAGATTAGAACTGGTAAACCAGAC	Mission Biotech
NbtraN_myc_F	GGCGAACAGAAGCTGATCAGCGAGGAAGATCTGTAATAACTGCAGTCCGGCAA	Mission Biotech
NbtraN_myc_R	CTGATCAGCTTCTGTTCGCCACCGCCGCTGCTCACGGTCACCTG	Mission Biotech
SL1344_TraN_F	TTATATTTTTGATGGTGCGACTAGAACAACAGAGGTCAGAACC	Mission Biotech
SL1344_TraN_R	CCCTTTTTTGCCGGACTGCATTATTACTTTACCTGCATCACCAGTGTCAGCGTGAACGTC	Mission Biotech
BamHI_TraN_surf_F	ACCATCATCACCACAGCCAGGATCCGACAACAGAGGTCAGAACC	Mission Biotech
NotI_TraN_surf_R	GTTCGACTTAAGCATTATGCGGCCGCTTACTTTACCTGCATCACCAGTGTCAGCGTGAACGTC	Mission Biotech
NdeI_NbOmpA/NbOmpC_F	CTTTAAGAAGGAGATATACATATGCAGGTGCAGCTGCAGGAAAG	Mission Biotech
XhoI_NbOmpA/NbOmpC_R	AGTGGTGGTGGTGGTGGTGCTCGAGGCTGCTCACGGTCACCTG	Mission Biotech
NbOmpA/NbOmpC_GFP_R1	TGCTCACGCTGCTGCTCACGGTCACCTG	Mission Biotech
NbOmpA/NbOmpC_GFP_F2	CGTGAGCAGCAGCGTGAGCAAGGGCGAG	Mission Biotech
XhoI_Nb-OmpA/Nb-OmpC_GFP_6His_R	AGTGGTGGTGGTGGTGGTGCTCGAGCTTGTACAGCTCGTCCATGCC	Mission Biotech

NdeI_NbtraN_F	GTATAAGAAGGAGATATACATATGCAGGTGCAGCTGCAGGAAAG	Mission Biotech
XhoI_NbtraN_R	CAGCGGTTTCTTTACCAGACTCGAGTTAGCTGCTCACGGTCAC	Mission Biotech
NbtraN_F47A/Y58A_F	GAACGCGAAGCCGTTGCCGGTATTACTGCTGGTAGTAGTACCGCTTATGCGGATAGCGTGAAAGGC	Mission Biotech
NbtraN_F47A/Y58A_R	ATCCGCATAAGCGGTACTACTACCAGCAGTAATACCGGCAACGGCTTCGCGTTCTTTGCCCGGCG	Mission Biotech
NbtraN_Y98A/L100A_F	TGCGCGGCTGCGCCGGCGGACGAAATCGGTTCTCATGAC	Mission Biotech
NbtraN_Y98A/L100A_R	GATTTCGTCCGCCGGCGCAGCCGCGCAATAATACACCG	Mission Biotech
NbtraN_S105C_F	GAAATCGGTTGCCATGACCCGCATTCTTATTGG	Mission Biotech
NbtraN_S105C_R	CGGGTCATGGCAACCGATTTCGTCCAGCGGGT	Mission Biotech
NbtraN_H106C_F	ATCGGTTCTTGCGACCCGCATTCTTATTGGGG	Mission Biotech
NbtraN_H106C_R	ATGCGGGTCGCAAGAACCGATTTCGTCCAGCG	Mission Biotech
NbcheckF	CTCTTGGGGCAAATAGTGCCAAA	Mission Biotech
NbcheckR	AGGAAGCCTGCATAACGCGAAG	Mission Biotech
pDSG_Nb1_seq_R	ACTACTAACGGTAACTTGCGTTC	Mission Biotech
pDSG_NbtraN_seq_R	ATGAGAACCGATTTCGTCCAGC	Mission Biotech
pBAD18_Seq_F	GCTATGCCATAGCATTTTTATCC	Mission Biotech
pBAD18_Seq_R	CTGATTTAATCTGTATCAGGCTG	Mission Biotech
pETDuet_MCS1_Seq_F	ATGCGTCCGGCGTAGA	Mission Biotech
pETDuet_MCS2_Seq_R	GCTAGTTATTGCTCAGCGG	Mission Biotech
pRE118 seq-F	GGTTGAGAAGCGGTGTAAGTG	Mission Biotech
pRE118 seq-R	GCAGGTCGATCCCAAGCTTC	Mission Biotech
MiSeq		

Nb_seq_UMI_offset_F1	TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNNCGGCAGCCTGCGCCT	Integrated DNA Technologies, Inc.
Nb_seq_UMI_offset_F2	TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNNNCGGCAGCCTGCGCCT	Integrated DNA Technologies, Inc.
Nb_seq_UMI_offset_F3	TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNNNNCGGCAGCCTGCGCCT	Integrated DNA Technologies, Inc.
Nb_seq_UMI_offset_F4	TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNNNNNACGGCAGCCTGCGCCT	Integrated DNA Technologies, Inc.
Nb_seq_UMI_offset_F5	TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNNNNNNACGGCAGCCTGCGCCT	Integrated DNA Technologies, Inc.
Nb_seq_offset_R1	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGCTGCTCACGGTCACC	Integrated DNA Technologies, Inc.
Nb_seq_offset_R2	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNGCTGCTCACGGTCACC	Integrated DNA Technologies, Inc.
Nb_seq_offset_R3	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNNGCTGCTCACGGTCACC	Integrated DNA Technologies, Inc.
Nb_seq_offset_R4	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNNNGCTGCTCACGGTCACC	Integrated DNA Technologies, Inc.
Nb_seq_offset_R5	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNNNNGCTGCTCACGGTCACC	Integrated DNA Technologies, Inc.

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